

PhotoCalib Tutorial: Robust Differential Photometry Pipeline

Overview

PhotoCalib is a sophisticated Python pipeline for performing differential photometry on astronomical image stacks. It's designed for time-series photometry applications like detecting exoplanet transits, monitoring variable stars, or tracking asteroid light curves. The pipeline automatically selects reference stars that are present in ALL images, ensuring consistent photometric calibration throughout your observing sequence.

Key Features

- **Automatic Reference Star Selection:** Intelligently selects non-saturated reference stars from the central region of your first image
- **Stack-Wide Verification:** Verifies each reference star exists in all images before using it
- **WCS-Based Tracking:** Uses World Coordinate System to track stars even when the field shifts between images
- **Robust Photometry:** Performs aperture photometry with background subtraction and error estimation
- **Diagnostic Outputs:** Creates detailed PDF reports showing aperture placement and radial profiles
- **Differential Photometry:** Measures target relative to multiple reference stars for high precision

Installation Requirements

```
pip install numpy pandas matplotlib astropy photutils scipy
# Optional but recommended for catalog matching:
pip install astroquery
```

Basic Usage

Example: Detecting an Exoplanet Transit

```
python3 photcalib_robust_fixed.py ~/Downloads/WASP-135b \
    -f "r'" \
    -o wasp135b_photometry_with_errors.csv \
    --target-ra "17:49:08.39" \
    --target-dec "+29:52:44.8" \
    --create-diagnostics \
    --verbose
```

This command will:

1. Process all FITS images in ~/Downloads/WASP-135b/
2. Use the r' (Sloan r-band) filter designation
3. Measure WASP-135b at the specified coordinates
4. Automatically find and verify 3 reference stars
5. Create diagnostic PDFs for each image
6. Output results to wasp135b_photometry_with_errors.csv

Command Line Arguments

Required Arguments

Argument	Description	Example
imagedir	Directory containing FITS images	~/Downloads/WASP-135b
-f, --filter	Photometric filter (Sloan system)	"r'", "g'", "i'"
-o, --output	Output CSV filename	photometry.csv
--target-ra	Target Right Ascension	"17:49:08.39" or 267.2849583
--target-dec	Target Declination	"+29:52:44.8" or 29.8791111

Optional Arguments

Argument	Default	Description
--fwhm	Auto	FWHM in pixels (auto-estimated if not provided)
--threshold-sigma	5.0	Detection threshold in sigma for finding stars
--max-standards	3	Maximum number of reference stars to use
--saturation	65000	Saturation level in ADU
--exptime-key	EXPTIME	FITS header keyword for exposure time
--jd-key	JD	FITS header keyword for Julian Date
--min-snr	20.0	Minimum SNR required for reference stars
--aperture-factor	1.5	Aperture radius as multiple of FWHM
--flux-ratio-limit	3.0	Max flux ratio between target and references
--search-radius	30.0	Search radius in pixels for finding stars
--center-fraction	0.5	Use central fraction of image for ref selection
--verification-sample	10	Number of images to sample for verification (0=all)
--create-diagnostics	Flag	Create diagnostic PDFs for each image
--verbose	Flag	Enable verbose output

How It Works

1. Reference Star Selection Phase

The pipeline begins by analyzing the first image in your stack:

```
=====
REFERENCE STAR SELECTION AND VERIFICATION
=====
```

1. Analyzing first image: WASP-135b-20250809at035709.fits
FWHM: 5.3 pixels
Target flux: 865654, SNR: 885.6
2. Selecting reference candidates from central 50% of image
Region: x=[1300, 3900], y=[975, 2925]
Found 47 potential reference stars

2. Verification Across Stack

Each candidate reference star is checked in sampled images throughout the stack:

3. Verifying candidates in 10 sampled images
Checking candidate 1: SNR=704.6, flux_ratio=0.66
✓ Verified in all sampled images
Checking candidate 2: SNR=635.9, flux_ratio=0.57
✓ Verified in all sampled images

3. Final Reference Selection

The best reference stars are selected based on:

- Signal-to-noise ratio (SNR)
- Flux similarity to target
- Central location in field
- Presence in all images

4. Selected 3 verified reference stars:
Reference 1:
Position: x=2566.2, y=1561.8
RA/Dec: 267.281234, 29.879456
SNR: 635.9, Flux ratio: 0.57

4. Processing All Images

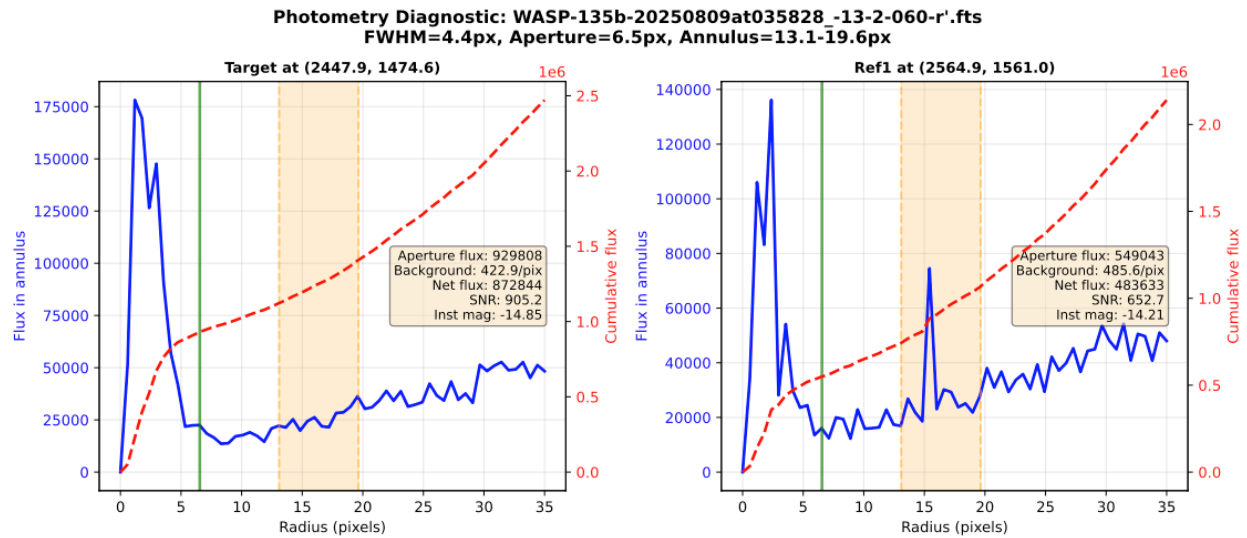
Each image is then processed using the verified reference stars:

```
Processing 120 images with 3 verified reference stars...
WASP-135b-20250809at035709.fits: 3/3 references, FWHM=5.3
Target: inst_mag=-14.843 (SATURATED), SNR=885.6
WASP-135b-20250809at035839.fits: 3/3 references, FWHM=5.2
Target: inst_mag=-14.847, SNR=892.1
```

Understanding Diagnostic Plots

When `--create-diagnostics` is enabled, the pipeline creates a 2-page PDF for each image:

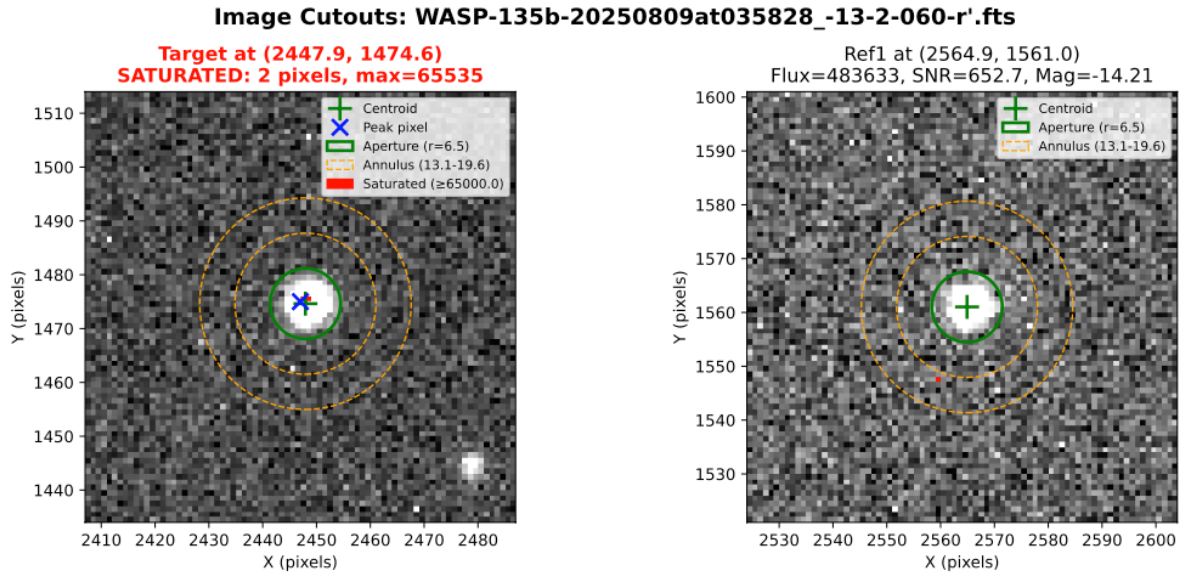
Page 1: Radial Profiles



Shows flux distribution around each star:

- **Green line:** Aperture boundary where flux is measured
- **Orange shading:** Background annulus region
- **Blue curve:** Flux in each annular ring
- **Red curve:** Cumulative flux from center
- **Text box:** Photometry measurements (flux, SNR, magnitude)

Page 2: Image Cutouts



Visual inspection of each star:

- **Green cross:** Refined centroid position
- **Blue X:** Peak pixel (if different from centroid)
- **Green circle:** Photometry aperture
- **Orange dashed circles:** Background annulus
- **Red pixels:** Saturated pixels (if any)
- **Title:** Position and photometry results

Output CSV File Structure

The output CSV contains one row per image with columns:

Basic Information

- `filename`: Image filename
- `jd`: Julian Date
- `filter`: Filter used
- `exptime`: Exposure time
- `fwhm`: Measured FWHM
- `n_sources`: Total sources measured (target + references)
- `n_standards`: Number of reference stars found

Target Measurements

- `target_x`, `target_y`: Pixel coordinates

- target_ra, target_dec: Sky coordinates
- target_inst_mag: Instrumental magnitude
- target_inst_mag_err: Magnitude error
- target_snr: Signal-to-noise ratio
- target_flux: Measured flux in ADU
- target_flux_err: Flux error
- target_saturated: Boolean saturation flag
- target_n_sat_pixels: Number of saturated pixels

Reference Star Measurements

For each reference star (std1, std2, std3):

- std1_x, std1_y: Pixel coordinates
- std1_inst_mag: Instrumental magnitude
- std1_inst_mag_err: Magnitude error
- std1_snr: Signal-to-noise ratio

Global Statistics

- target_inst_mag_stddev_all: Standard deviation of target magnitude across all images
- error: Any error messages

Practical Examples

Example 1: Exoplanet Transit Detection

Monitor WASP-135b for a planetary transit:

```
python3 photcalib_robust_fixed.py ~/data/WASP-135b \
  -f "r" \
  -o wasp135b_transit.csv \
  --target-ra "17:49:08.39" \
  --target-dec "+29:52:44.8" \
  --max-standards 5 \
  --min-snr 30 \
  --create-diagnostics
```

Example 2: Variable Star Monitoring

Track a variable star with custom aperture settings:

```
python3 photcalib_robust_fixed.py ~/data/RR_Lyrae \
  -f "g" \
  -o rr_lyrae_lightcurve.csv \
  --target-ra "19:25:27.91" \
```

```
--target-dec "+42:47:03.7" \  
--aperture-factor 2.0 \  
--fwhm 4.5
```

Example 3: Asteroid Light Curve

Measure a moving asteroid (update coordinates for each night):

```
python3 photcalib_robust_fixed.py ~/data/asteroid_433 \  
-f "r" \  
-o eros_rotation.csv \  
--target-ra "03:15:42.3" \  
--target-dec "+15:23:18" \  
--search-radius 50 \  
--verification-sample 0 # Check all images
```

Troubleshooting

"No suitable reference candidates found"

- Increase `--center-fraction` to search larger area (e.g., 0.7)
- Decrease `--min-snr` requirement (e.g., 15)
- Increase `--flux-ratio-limit` to allow dimmer/brighter references

"Could not find reference in this image"

- Field may have shifted significantly
- Increase `--search-radius` (e.g., 50 pixels)
- Check if images have valid WCS headers

Target shows as SATURATED

- Reduce exposure time for future observations
- Consider using a narrowband filter or neutral density filter
- Saturation makes photometry unreliable

Large scatter in target magnitudes

- Check diagnostic PDFs for clouds or tracking issues
- Verify reference stars are stable (not variables)
- Consider weather conditions during observations

Processing Pipeline Summary

The complete workflow looks like:

1. Load first image → Estimate FWHM → Find target
↓
2. Detect all sources → Select candidates from center
↓
3. Verify each candidate exists in sampled images
↓
4. Choose best reference stars (up to max-standards)
↓
5. Process each image:
 - Find target using WCS
 - Find each reference using WCS
 - Perform aperture photometry
 - Create diagnostics (optional)↓
6. Output CSV with all measurements

Tips for Best Results

1. **Image Quality:** Ensure images are well-focused and properly calibrated (bias/dark/flat)
2. **Reference Star Selection:** The pipeline automatically selects the best references, but you can tune:
 - `--center-fraction`: Smaller values (0.3) focus on very central stars
 - `--flux-ratio-limit`: Lower values (2.0) select references closer in brightness to target
3. **Aperture Optimization:**
 - Default `--aperture-factor 1.5` works well for most seeing conditions
 - Increase to 2.0-2.5 for poor seeing
 - Decrease to 1.0-1.2 for excellent seeing and faint targets
4. **Verification Strategy:**
 - `--verification-sample 10` is faster but may miss problem images
 - `--verification-sample 0` checks all images (slower but thorough)
5. **Diagnostic Review:** Always check a few diagnostic PDFs to verify:
 - Apertures are well-placed
 - No contaminating stars in apertures
 - Background regions are source-free
 - Target isn't saturated

Data Analysis

After running PhotoCalib, analyze the output CSV:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load photometry
df = pd.read_csv('wasp135b_photometry.csv')

# Plot light curve
plt.figure(figsize=(12, 6))
plt.scatter(df['jd'], df['target_inst_mag'],
```



```

        c=df['target_snr'], cmap='viridis', s=20)
plt.xlabel('Julian Date')
plt.ylabel('Instrumental Magnitude')
plt.gca().invert_yaxis() # Brighter = lower magnitude
plt.colorbar(label='SNR')
plt.title('WASP-135b Light Curve')
plt.show()

# Check for transit
transit_depth = df['target_inst_mag'].max() - df['target_inst_mag'].min()
print(f"Light curve depth: {transit_depth:.3f} magnitudes")

```

Conclusion

PhotoCalib provides professional-grade differential photometry with minimal user intervention. Its automatic reference star verification ensures consistent measurements across your entire observing run, making it ideal for time-series photometry of exoplanet transits, variable stars, and solar system objects.